

1. Image Classification Extension

Problem Statement

In real-world scenarios, image datasets vary significantly in size, quality, and domain. A fixed model often fails to generalize well. Hence, there is a need for a **customizable image classification system** that can adapt to different datasets while maintaining good accuracy.

Approach

- Built an image classification extension using **TensorFlow and PyTorch**
- Leveraged **transfer learning** with pre-trained models such as:
 - ResNet
 - AlexNet
 - VGG
 - SqueezeNet
 - DenseNet
 - Inception
 - Vision Transformers (ViT, DeiT via Hugging Face)

Fine-Tuning

- Dataset organized in class-wise folders
- Automatic **75% training / 25% validation split**
- Customizable parameters:
 - Number of layers
 - Batch size
 - Learning rate
 - Optimizer
 - Epochs
- Used **data augmentation, class-weighted learning, and callbacks** like early stopping and learning-rate scheduling

Inference

- Trained model exported to **ONNX**
- Images are preprocessed and passed to the model
- Output includes:
 - Predicted class label
 - Confidence score
- Images are automatically placed into output folders based on predicted class

Results

- Animal classification: ~87% accuracy
- Facial emotion recognition: ~80% accuracy
- Flower dataset: ~94% accuracy